## Discovering Matplotlib Basics

### *Section 1: Learn*

### What is Matplotlib?

Matplotlib is a **popular Python library for data visualization**.

It helps in creating **charts, graphs, and plots** to better understand data patterns.

### Why is Matplotlib Important?

- **Improves Data Understanding** – Converts raw data into easy-to-read visuals.
- **Customizable & Flexible** – Offers multiple plotting styles and customization options.
- **Works with Other Libraries** – Integrates well with **Pandas, NumPy, and Seaborn**.

### How to Use Matplotlib?

1. **Import Matplotlib** – Use import matplotlib.pyplot as plt.
2. **Create a Plot** – Use functions like plot(), bar(), hist(), etc.
3. **Customize the Plot** – Add labels, titles, legends, and colors.
4. **Display the Plot** – Use plt.show() to visualize.

### Common Types of Plots in Matplotlib

- **Line Plot** – Shows trends over time.
- **Bar Chart** – Compares categorical data.
- **Histogram** – Displays the distribution of numerical data.
- **Scatter Plot** – Shows relationships between two variables.

**Real-Life Applications of Matplotlib**

- **Stock Market Analysis** – Visualizing stock price trends.
- **Sports Analytics** – Analyzing player performance.
- **Weather Forecasting** – Plotting temperature variations.

**Anecdote: How NASA Used Matplotlib to Study Climate Change**

NASA scientists use **Matplotlib** to visualize **global temperature changes over decades**.

By analyzing these trends, they provide **valuable insights into climate change and its effects**.

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***Section 2: Practice***

**1. Creating a Simple Line Plot**

```
import matplotlib.pyplot as plt


# Data points
years = [2015, 2016, 2017, 2018, 2019, 2020]
sales = [100, 150, 200, 250, 300, 400]


# Creating the plot
plt.plot(years, sales, marker='o', linestyle='-', color='b', label="Sales Growth")


# Adding labels and title
plt.xlabel("Year")
plt.ylabel("Sales in Lakhs")
plt.title("Company Sales Over Years")
```

```
# Displaying the legend
plt.legend()

# Showing the plot
plt.show()
```

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**2. Creating a Bar Chart**

```
# Categories and values
products = ["Product A", "Product B", "Product C"]
sales = [5000, 7000, 6000]

# Creating a bar chart
plt.bar(products, sales, color=['blue', 'red', 'green'])

# Adding labels and title
plt.xlabel("Products")
plt.ylabel("Sales")
plt.title("Sales Comparison")

# Displaying the plot
plt.show()
```

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### 3. Creating a Histogram

```
import numpy as np


# Generating random data
data = np.random.normal(50, 10, 100)


# Creating a histogram
plt.hist(data, bins=10, color='purple', edgecolor='black')


# Adding labels and title
plt.xlabel("Value")
plt.ylabel("Frequency")
plt.title("Histogram of Random Data")


# Displaying the plot
plt.show()
```

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### 4. Creating a Scatter Plot

```
# Sample data
x = [10, 20, 30, 40, 50]
y = [5, 15, 25, 35, 45]


# Creating a scatter plot
plt.scatter(x, y, color='red', marker='o')
```

```
# Adding labels and title
plt.xlabel("X-axis")
plt.ylabel("Y-axis")
plt.title("Scatter Plot Example")

# Displaying the plot
plt.show()
```

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***Section 3: Know More***

**Frequently Asked Questions (FAQs)**

**1. How is Matplotlib different from Seaborn?**

- **Matplotlib** – Low-level, highly customizable.
- **Seaborn** – Built on Matplotlib, provides more advanced visualizations with simpler syntax.

**2. Why do we use plt.show()?**

It **displays the plot** on the screen after defining it.

**3. How can I change the figure size in Matplotlib?**

Use plt.figure(figsize=(width, height)) before creating the plot.

**4. How do I save a Matplotlib plot?**

Use plt.savefig("filename.png") to save the figure.

**5. What are subplots in Matplotlib?**

Subplots allow **multiple plots in one figure** using plt.subplot(rows, cols, index).

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**Conclusion:**

Matplotlib is an essential tool for **data visualization in Python**.

By mastering **basic plots and customization**, we can create **insightful visual representations** of data for analysis and decision-making.